

Selectime: Validation-selected inputs and frozen forecast mixtures

1 Candidate family

Given an observed target prefix and a horizon H , compare independent univariate forecasting, joint multivariate forecasting, past-only target augmentation with $\sqrt{|x|}$ and $\text{sign}(x)$, and retrieved covariates for $K \in \{1, 5, 10, 15, 20\}$. The default backbone is frozen Chronos-2; TS-ICL omits native multivariate inputs and the two scope controls. Chronos-Bolt compares univariate vanilla with direct-horizon mixing only. Native backbone forecasts are medians. There is no backbone parameter fitting.

2 Aligned retrieval

For a fixed retrieval lookback r , represent each query and neighbor as $z(x) = (x - \mu(x)) / \max(\sigma(x), 10^{-8})$, using only that lookback’s finite past values. Rank eligible neighbors by Euclidean distance on finite overlap (at least 80%, with squared distances adjusted for overlap). Require a complete maximum- K list for covariate candidates; smaller K values reuse prefixes. Direct-horizon transfer searches independently for one eligible neighbor. The datastore is within one dataset/frequency task, across items and variates by default. A same-series ablation is retained. Task-specific calendar phases and frequency-dependent strides define the eligible dates.

Before supplying a neighbor trajectory s_i to the backbone, apply

$$\tilde{s}_i = (s_i - \mu_i) \frac{\max(\sigma_q, 10^{-8})}{\max(\sigma_i, 10^{-8})} + \mu_q,$$

where both sets of statistics use past retrieval lookbacks only. Supply the rescaled neighbor past and observed future as covariates. Self-augmentation has no future covariates.

3 Chronology and selection

The validation datastore ends before validation starts. The validation period immediately precedes the official TIME test interval. After freezing method and K , expand retrieval to all eligible pre-test history. Complete neighbor futures must end before the applicable boundary and be observed before the query; no test observation enters the datastore.

Score candidates by mean squared error divided by the mean seasonal squared difference over the observed target prefix (MSSE). Average variates within each validation date before comparing dates for task-level selection; per-variate selection scores each item/variant’s dates independently. Use Adapttime’s paired moving-date-block bootstrap: a candidate is admissible when its mean loss difference from the observed best is at most one bootstrap standard error. Prefer univariate, multivariate, self-augmentation, then smaller K within that threshold. Choices remain fixed throughout testing.

4 Binary scope control and soft mixtures

The scope selector applies the same task bootstrap rule to only univariate and multivariate vanilla. It is a hard gate. The scope mixture blends those forecasts; the top-5 mixture blends univariate with the top-5 covariate forecast. For direct-horizon mixing, the alternative is the nearest historical neighbor’s rescaled continuation $\tilde{s}_{1,\text{future}}$, without invoking the backbone on that neighbor or passing it as a covariate.

For each soft mixture, compare the two predictions by MSSE on eligible validation item/variant/window rows. Exclude alternative fallback rows and non-finite losses. Let n be the number of paired rows and w the number of alternative wins, with ties contributing one half. Adaption’s Beta(1, 1) win-frequency rule gives

$$p = \frac{1 + w}{2 + n}, \quad \hat{y} = (1 - p)\hat{y}_{\text{uni}} + p\hat{y}_{\text{alternative}}.$$

With no usable rows set $p = 0$. Freeze one scalar p per task before test. An all-missing available target history is not sent to a backbone and contributes no validation loss. If no usable validation rows remain, the canonical univariate method is selected. Official test support is unchanged. These controls do not enter the original task/per-variant selectors. Scope controls exist only for Chronos-2; top-5 mixing also exists for TS-ICL; direct-horizon mixing exists for all three backbones.

5 Evidence

Official test windows and the Seasonal Naive metric-cell grid remain unchanged. Unavailable retrieval and non-finite candidate cells use canonical univariate forecasts with recorded masks. The initial $K \leq 15$ Chronos-2 study is analyzed in the results recap. The expanded Chronos-2/Bolt comparison is analyzed separately. TS-ICL remains unevaluated under the finite-history validation contract.